

Problem 3.19

Given $i_5 = 0.1$ and $A(4) = 146.41$. Find $A(5)$.

Solution

By definition,

$$i_5 = \frac{A(5) - A(4)}{A(4)}.$$

Substituting the given values,

$$0.1 = \frac{A(5) - A(4)}{A(4)}.$$

Multiplying both sides by $A(4)$,

$$(0.1)A(4) = A(5) - A(4).$$

Therefore,

$$0.1(146.41) = A(5) - 146.41.$$

So,

$$0.1(146.41) + 146.41 = A(5).$$

Hence,

$$161.051 = A(5).$$